

Project Report: Hierarchical Clustering of Gene Expression Data

1. Introduction

- **Objective:** To identify meaningful biological groupings within leukemia gene expression data using unsupervised machine learning.
 - **Technique Used:** Hierarchical Clustering, a method based on computing pairwise distances to form a tree-like structure (dendrogram) showing data similarity.
 - **Application:** Helps in identifying patterns, discovering disease subtypes, and visualizing the gene expression structure.
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2. Dataset Description

- **Source:** Leukemia_GSE9476 gene expression dataset.
- **Features:** Thousands of gene expression values per sample.
- **Metadata Columns:** samples, type (e.g., AML, Bone Marrow, etc.)

Initial Shape: (n_samples, n_features + 2)

Preprocessing:

- Dropped metadata columns.
 - Converted all values to numeric using `pd.to_numeric(errors='coerce')`.
 - Handled missing values.
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3. Libraries and Functions Used

Python Libraries:

- **pandas (pd):**
 - `read_csv()`: To load CSV files.
 - `drop()`: To remove non-numeric columns.
 - `apply()`: Used to convert data types.

- `groupby()`: For aggregating gene expression values per cluster.
- **numpy (np):**
 - Core numerical operations and array handling.
- **matplotlib.pyplot (plt):**
 - `figure()`: Initializes the plot.
 - `show()`: Displays the final plot.
 - `tight_layout()`: Adjusts subplot spacing.
 - `xlabel()`, `ylabel()`, `title()`: For labelling plots.
- **seaborn (sns):**
 - `heatmap()`: Creates a color-coded matrix visualization for average gene expression.
- **scipy.cluster.hierarchy:**
 - `linkage()`: Computes the hierarchical clustering.
 - `dendrogram()`: Plots the dendrogram showing the hierarchy.
 - `fcluster()`: Extracts flat clusters from hierarchical clustering results.
- **sklearn.preprocessing.StandardScaler:**
 - `fit_transform()`: Standardizes features by removing the mean and scaling to unit variance, crucial for distance-based clustering algorithms.

These libraries provide an efficient toolkit for performing, visualizing, and analyzing hierarchical clustering on high-dimensional biological data. Each function plays a vital role from data preparation to the interpretation of clustering results.

4. Methodology

Step-by-Step Breakdown:

1. **Data Loading:**
 - Loaded using `pd.read_csv()`
2. **Preprocessing:**
 - Removed samples and type columns.
 - Applied `StandardScaler` to normalize features.

3. Clustering:

- Used `linkage(scaled_data, method='ward')`.
- Ward's method minimizes the variance within clusters.

4. Visualization - Dendrogram:

- Generated using `dendrogram()`
- Displays hierarchical relationships.

Dendrogram Insights:

- Reveals clear branching of the data into clusters.
- Helps in estimating the number of distinct clusters by looking at the longest vertical distances that don't cross any horizontal link.
- Groupings show samples of similar gene expression clustered together, especially similar biological conditions such as AML.

5. Cluster Assignment:

- Extracted 4 clusters using `fcluster()`.

6. Cluster-wise Expression Analysis:

- Grouped by cluster, averaged gene expressions.
- Visualized using `sns.heatmap()`.

Heatmap Insights:

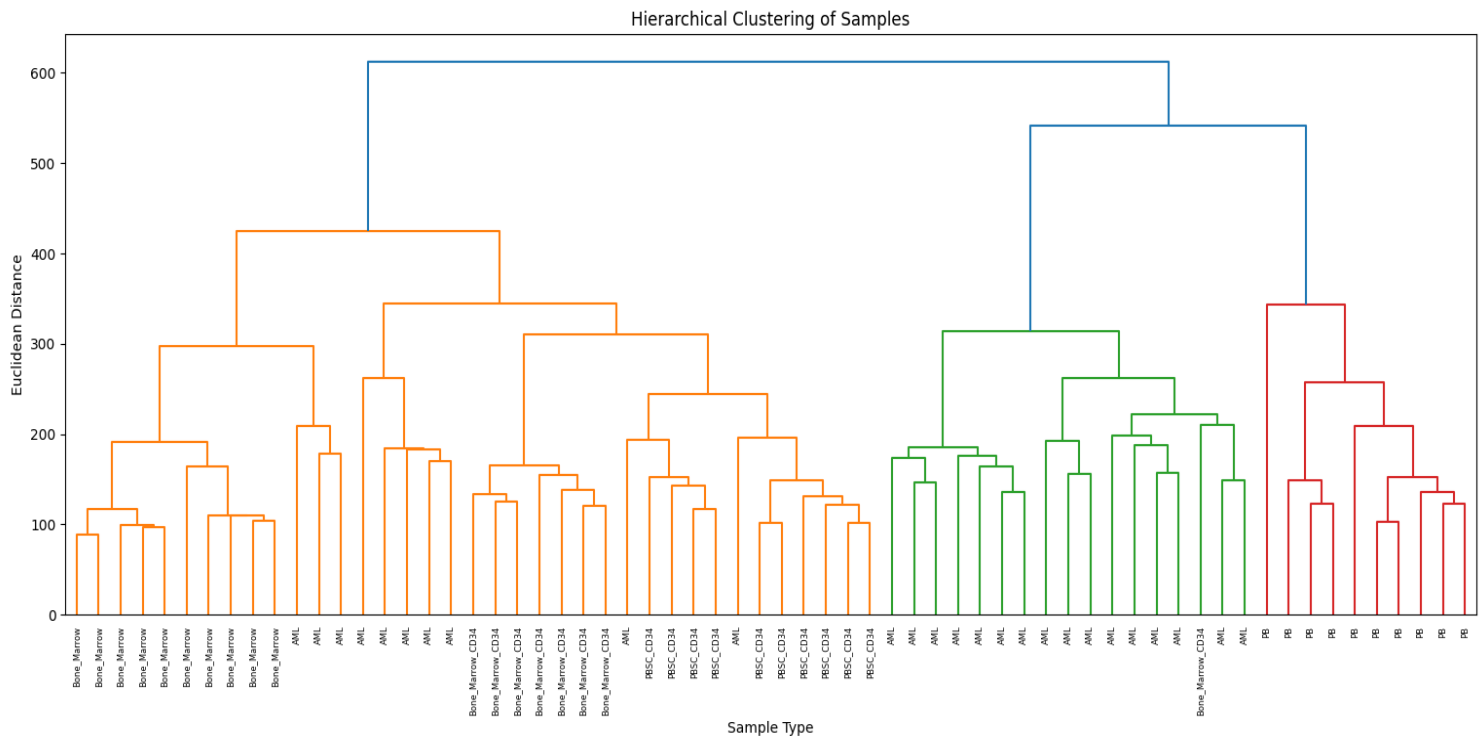
- Each column represents a cluster; each row a gene.
- Color intensity indicates level of gene expression (e.g., yellow = high, blue = low in Viridis color scheme).
- Enables quick detection of overexpressed or underexpressed genes in clusters.
- Useful in identifying potential biomarkers and understanding cluster-specific gene activity patterns.

5. Insights from the Results

• Dendrogram Observations:

- Samples grouped based on expression similarity.
- Some AML and Bone Marrow samples cluster closely, suggesting expression similarities.

Clear hierarchy indicates possible biological relationships among sample types.

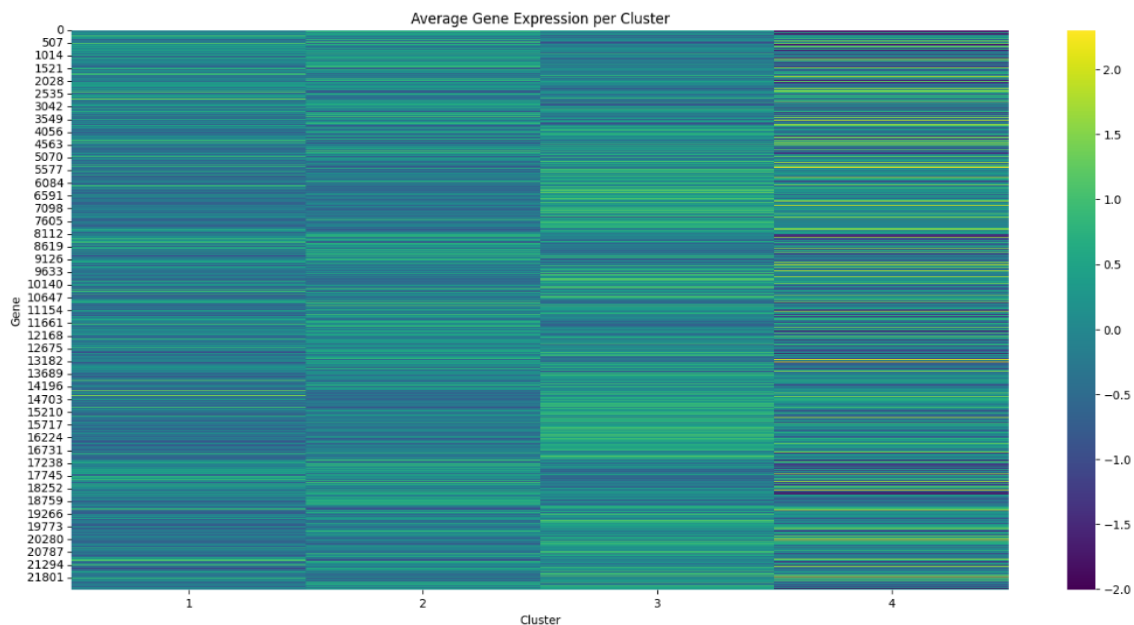


- **Cluster Assignments:**

- Clear separation of samples into 4 meaningful groups.
- Some overlap between similar biological types indicates shared expression patterns.

- **Heatmap Analysis:**

- Distinct gene expression profiles for each cluster.
- Certain genes are upregulated in specific clusters, which may be potential biomarkers.



- Provides a visual summary of how genes behave across different clusters.
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6. Terminologies & Concepts

- **Standardization:** Rescaling data to have zero mean and unit variance.
 - **Euclidean Distance:** Measure of dissimilarity used in clustering.
 - **Dendrogram:** Tree diagram showing arrangement of clusters.
 - **Heatmap:** Visual representation of gene expression values.
 - **Cluster:** Group of samples with similar gene expression patterns.
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7. Advantages of Hierarchical Clustering

- No need to predefine the number of clusters.
 - Provides a tree (dendrogram) for visual assessment.
 - Effective for gene expression data due to biological similarity.
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8. Limitations

- Sensitive to noise and outliers.
 - Computationally intensive for very large datasets.
 - Fixed hierarchical structure once created (no re-clustering).
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9. Future Scope & Improvements

- **Biological Validation:** Perform gene ontology or pathway enrichment analysis.
 - **Alternative Linkage Methods:** Try complete, average, single for comparison.
 - **Dimensionality Reduction:** Apply PCA or t-SNE for better visualization.
 - **Model Comparison:** Benchmark with KMeans, DBSCAN.
 - **Automation:** Create a pipeline to preprocess and cluster new gene data automatically.
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10. Conclusion

- Hierarchical clustering successfully grouped gene expression profiles without prior labels.
 - The approach revealed biologically relevant patterns and potential subtypes.
 - Visualization via dendrogram and heatmap provided insights into structure and expression variations.
 - This model can aid in early diagnosis, treatment planning, and research into disease subtypes.
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11. Sample Code Overview (from Clustering.py)

```
from scipy.cluster.hierarchy import linkage, dendrogram, fcluster
from sklearn.preprocessing import StandardScaler
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

# Preprocess
df = pd.read_csv('Leukemia_GSE9476.csv')
data = df.drop(columns=['samples', 'type'])
scaled_data = StandardScaler().fit_transform(data)

# Clustering
linked = linkage(scaled_data, method='ward')
cluster_labels = fcluster(linked, t=4, criterion='maxclust')

# Visualization
plt.figure(figsize=(15,6))
dendrogram(linked)
```

```
plt.show()

# Heatmap
df['Cluster'] = cluster_labels
cluster_mean = pd.DataFrame(scaled_data).groupby(df['Cluster']).mean().T
sns.heatmap(cluster_mean, cmap='viridis')
plt.show()
```